Practical-3: Time Series Decomposition

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PRACTICAL-3: Time Series Decomposition

Question:

Consider the following datasets from the R library and write R code to decompose the time-series data into its trend, seasonal, and residual components. Furthermore, identify the dominating component(s) in the dataset: (a) Nottem data (b) AirPassengers data

Understanding/Analysis Approach:

1. **Objective**: Analyze time series datasets to extract trend, seasonal, and residual components
2. **Datasets**:
   - Nottem: Monthly temperatures at Nottingham (1920-1939)
   - AirPassengers: Monthly airline passenger counts (1949-1960)
3. **Method**: Use classical decomposition to separate components
4. **Analysis**: Calculate variance of each component to identify dominant pattern
5. **Comparison**: Contrast results between climate data (Nottem) and economic data (AirPassengers)

Complete R Code:

```
# ========== PRACTICAL-3: Time Series Decomposition ==========
# Load required libraries
library(ggplot2)
library(forecast)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
library(ggfortify)
```

```
## Registered S3 methods overwritten by 'ggfortify':
##   method                 from
```

```
##   autoplot.Arima         forecast
##   autoplot.acf           forecast
##   autoplot.ar            forecast
##   autoplot.bats          forecast
##   autoplot.decomposed.ts forecast
##   autoplot.ets           forecast
##   autoplot.forecast      forecast
##   autoplot.stl           forecast
##   autoplot.ts            forecast
##   fitted.ar              forecast
##   fortify.ts             forecast
##   residuals.ar           forecast
```

```r
cat("PRACTICAL-3: TIME SERIES DECOMPOSITION\n")
```

```
## PRACTICAL-3: TIME SERIES DECOMPOSITION
```

```r
cat(rep("=", 60), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
# ------------------------------------------------------------
# (a) NOTTEM DATA ANALYSIS
# ------------------------------------------------------------
cat("\n", rep("-", 50), "\n", sep="")
```

```
##
## --------------------------------------------------
```

```r
cat("(a) NOTTEM DATASET ANALYSIS\n")
```

```
## (a) NOTTEM DATASET ANALYSIS
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# Load Nottem dataset
data("nottem")
cat("1. Loading Nottem dataset...\n")
```

```
## 1. Loading Nottem dataset...
```

```r
cat("   - Time period:", start(nottem), "to", end(nottem), "\n")
```

```
##    - Time period: 1920 1 to 1939 12
```

```r
cat("   - Frequency:", frequency(nottem), "(monthly data)\n")
```

```
##    - Frequency: 12 (monthly data)
```

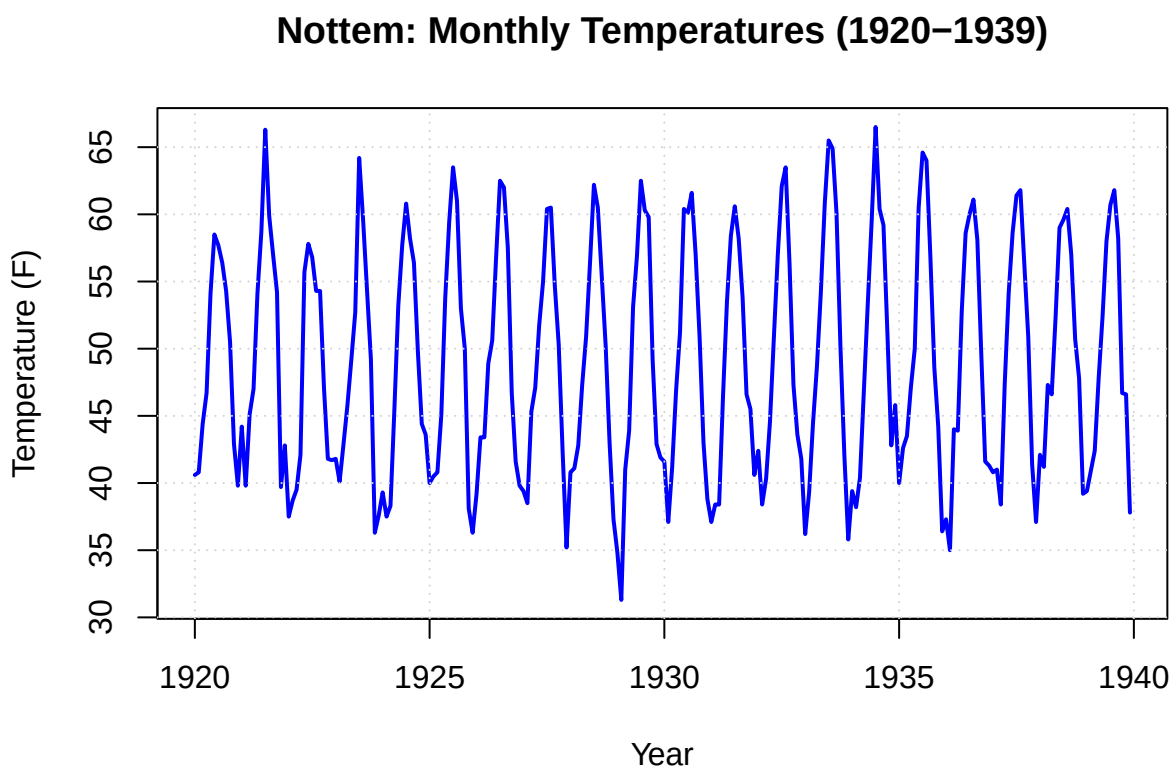
```r
cat("   - Observations:", length(nottem), "\n\n")
```

```
##    - Observations: 240
```

```r
# Plot original data
cat("2. Plotting original time series...\n")
```

```
## 2. Plotting original time series...
```

```r
plot(nottem,
     main = "Nottem: Monthly Temperatures (1920-1939)",
     ylab = "Temperature (F)",
     xlab = "Year",
     col = "blue",
     lwd = 2)
grid()
```



```r
# Decompose time series
cat("3. Decomposing time series...\n")
```

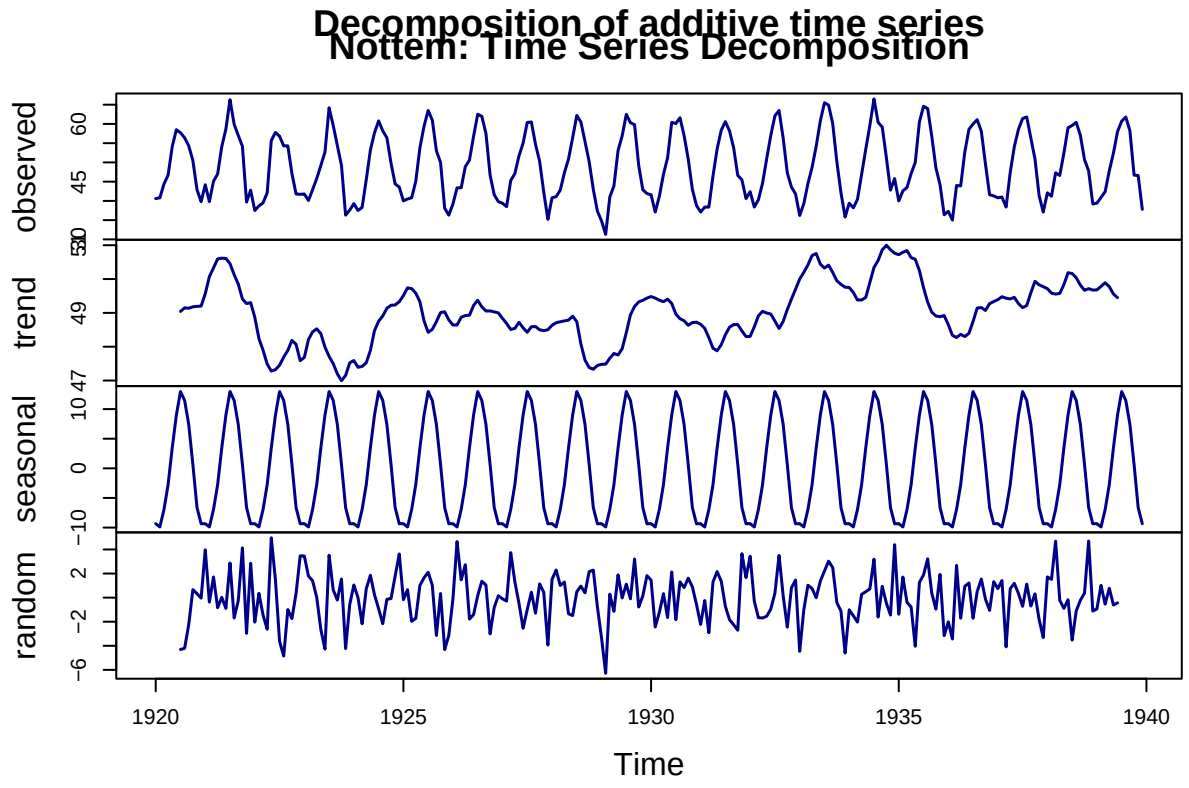
```
## 3. Decomposing time series...
```

```r
nottem_decomp <- decompose(nottem)

# Plot decomposition
cat("4. Plotting decomposition components...\n")
```

```
## 4. Plotting decomposition components...
```

```r
plot(nottem_decomp,
     col = "darkblue",
     lwd = 1.5)
title(main = "Nottem: Time Series Decomposition")
```




```r
# Calculate variances
cat("5. Calculating component variances...\n")
```

```
## 5. Calculating component variances...
```

```r
nottem_var_total <- var(nottem, na.rm = TRUE)
nottem_var_trend <- var(nottem_decomp$trend, na.rm = TRUE)
nottem_var_seasonal <- var(nottem_decomp$seasonal, na.rm = TRUE)
nottem_var_residual <- var(nottem_decomp$random, na.rm = TRUE)

# Display results
cat("\n6. VARIANCE ANALYSIS (Nottem):\n")
```

```
##
## 6. VARIANCE ANALYSIS (Nottem):
```

```r
cat("   Total variance:", round(nottem_var_total, 3), "\n")
```

```
##    Total variance: 73.485
```

```r
cat("   Trend variance:", round(nottem_var_trend, 3), "\n")
```

```
##    Trend variance: 0.781
```

```r
cat("   Seasonal variance:", round(nottem_var_seasonal, 3), "\n")
```

```
##    Seasonal variance: 68.605
```

```r
cat("   Residual variance:", round(nottem_var_residual, 3), "\n\n")
```

```
##    Residual variance: 4.322
```

```r
cat("7. PERCENTAGE OF VARIANCE EXPLAINED:\n")
```

```
## 7. PERCENTAGE OF VARIANCE EXPLAINED:
```

```r
cat("   Trend:", round((nottem_var_trend/nottem_var_total)*100, 1), "%\n")
```

```
##    Trend: 1.1 %
```

```r
cat("   Seasonal:", round((nottem_var_seasonal/nottem_var_total)*100, 1), "%\n")
```

```
##    Seasonal: 93.4 %
```

```r
cat("   Residual:", round((nottem_var_residual/nottem_var_total)*100, 1), "%\n\n")
```

```
##    Residual: 5.9 %
```

```r
# Determine dominating component
cat("8. DOMINATING COMPONENT IDENTIFICATION:\n")
```

```
## 8. DOMINATING COMPONENT IDENTIFICATION:
```

```r
if (nottem_var_seasonal > nottem_var_trend && nottem_var_seasonal > nottem_var_residual) {
  cat("   DOMINATING COMPONENT: SEASONAL\n")
  cat("   Reason: Seasonal variance (", round((nottem_var_seasonal/nottem_var_total)*100, 1), "%) is hig
} else if (nottem_var_trend > nottem_var_seasonal && nottem_var_trend > nottem_var_residual) {
  cat("   DOMINATING COMPONENT: TREND\n")
} else {
  cat("   DOMINATING COMPONENT: RANDOM/RESIDUAL\n")
}
```

```
##    DOMINATING COMPONENT: SEASONAL
##    Reason: Seasonal variance (93.4%) is highest
```

```r
# ------------------------------------------------------------
# (b) AIRPASSENGERS DATA ANALYSIS
# ------------------------------------------------------------
cat("\n\n", rep("-", 50), "\n", sep="")
```

```
##
##
## --------------------------------------------------
```

```r
cat("(b) AIRPASSENGERS DATASET ANALYSIS\n")
```

```
## (b) AIRPASSENGERS DATASET ANALYSIS
```

```r
cat(rep("-", 50), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
# Load AirPassengers dataset
data("AirPassengers")
cat("1. Loading AirPassengers dataset...\n")
```

```
## 1. Loading AirPassengers dataset...
```

```r
cat("   - Time period:", start(AirPassengers), "to", end(AirPassengers), "\n")
```

```
##    - Time period: 1949 1 to 1960 12
```

```r
cat("   - Frequency:", frequency(AirPassengers), "(monthly data)\n")
```

```
##    - Frequency: 12 (monthly data)
```

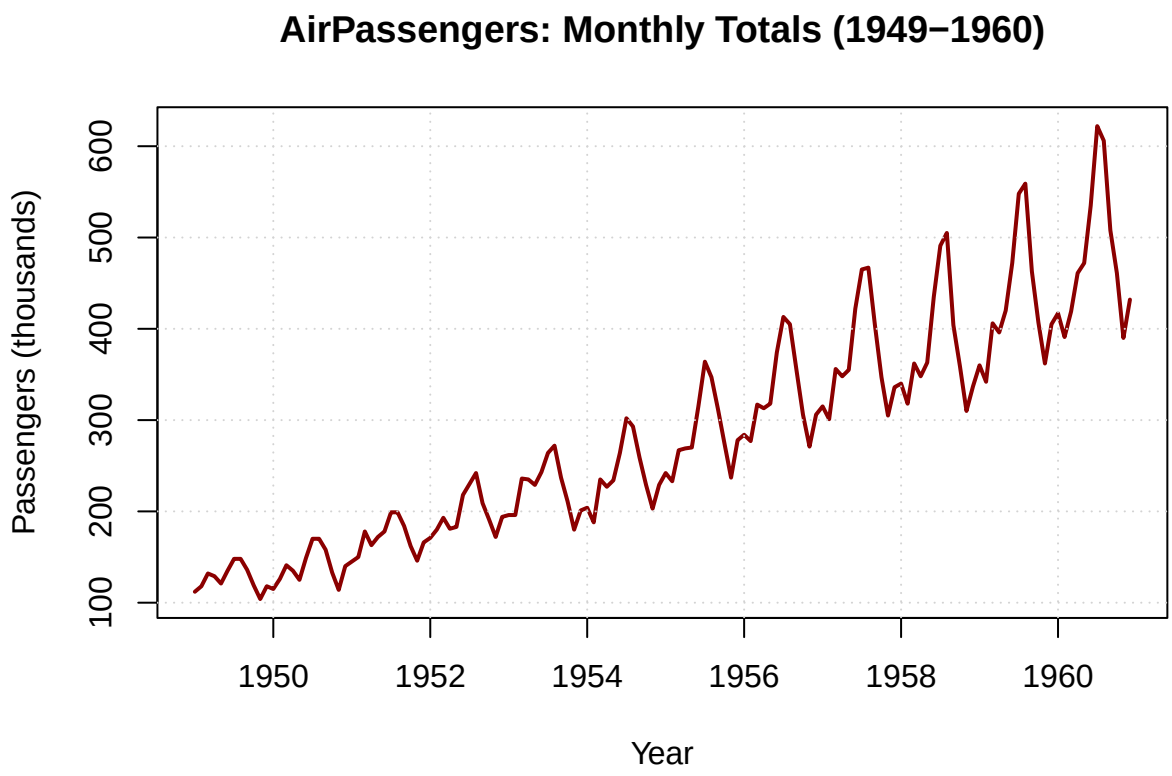
```r
cat("   - Observations:", length(AirPassengers), "\n\n")
```

```
##    - Observations: 144
```

```r
# Plot original data
cat("2. Plotting original time series...\n")
```

```
## 2. Plotting original time series...
```

```r
plot(AirPassengers,
     main = "AirPassengers: Monthly Totals (1949-1960)",
     ylab = "Passengers (thousands)",
     xlab = "Year",
     col = "darkred",
     lwd = 2)
grid()
```




```r
# Decompose time series
cat("3. Decomposing time series...\n")
```

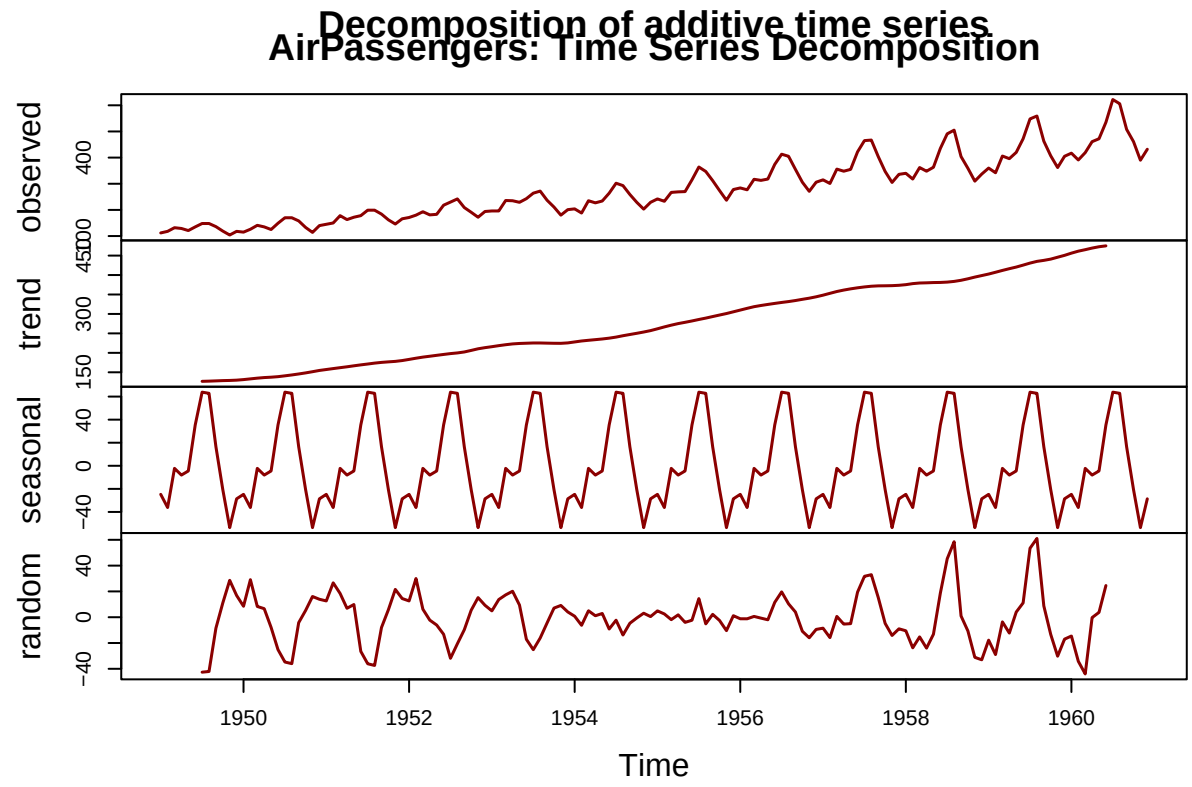
```
## 3. Decomposing time series...
```

```r
air_decomp <- decompose(AirPassengers)

# Plot decomposition
cat("4. Plotting decomposition components...\n")
```

```
## 4. Plotting decomposition components...
```

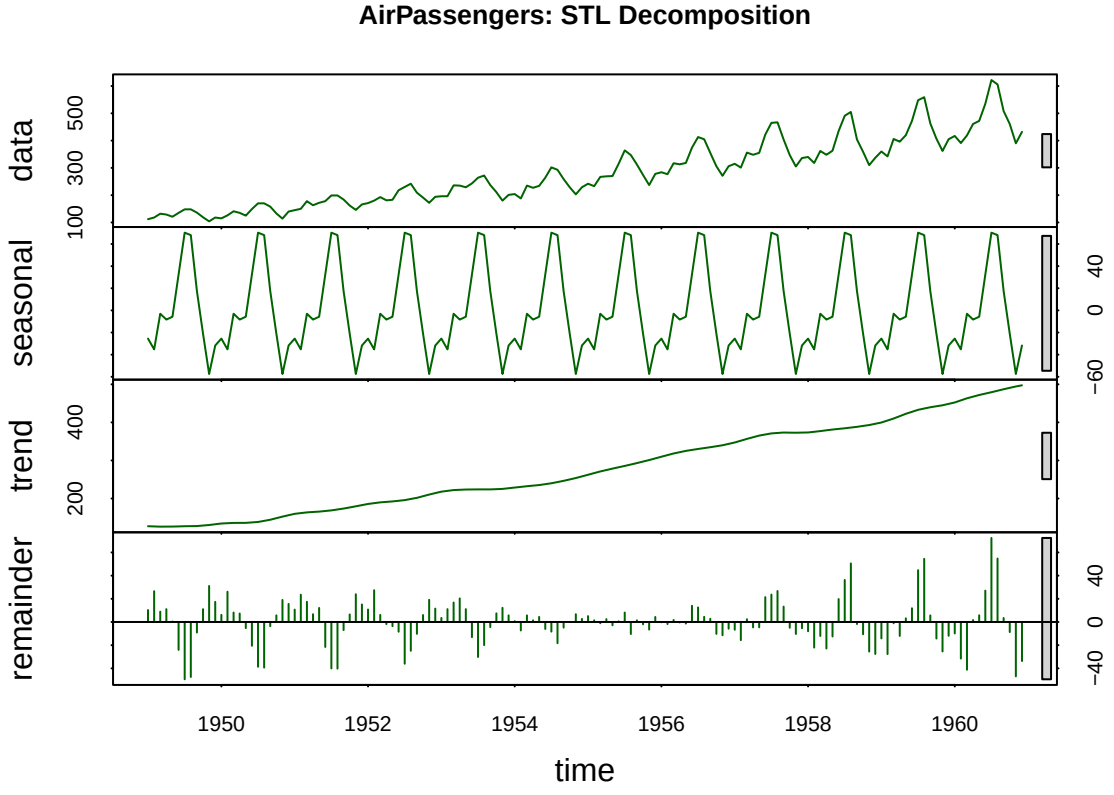
```r
plot(air_decomp,
     col = "darkred",
     lwd = 1.5)
title(main = "AirPassengers: Time Series Decomposition")
```




```r
# Alternative STL decomposition
cat("5. Performing STL decomposition (alternative method)...\n")
```

```
## 5. Performing STL decomposition (alternative method)...
```

```r
air_stl <- stl(AirPassengers, s.window = "periodic")
plot(air_stl,
     main = "AirPassengers: STL Decomposition",
     col = "darkgreen")
```




```
# Calculate variances
cat("6. Calculating component variances...\n")
```

```
## 6. Calculating component variances...
```

```
air_var_total <- var(AirPassengers, na.rm = TRUE)
air_var_trend <- var(air_decomp$trend, na.rm = TRUE)
air_var_seasonal <- var(air_decomp$seasonal, na.rm = TRUE)
air_var_residual <- var(air_decomp$random, na.rm = TRUE)

# Display results
cat("\n7. VARIANCE ANALYSIS (AirPassengers):\n")
```

```
##
## 7. VARIANCE ANALYSIS (AirPassengers):
```

```
cat("   Total variance:", round(air_var_total, 3), "\n")
```

```
##    Total variance: 14391.92
```

```
cat("   Trend variance:", round(air_var_trend, 3), "\n")
```

```
##    Trend variance: 10524.94
```

```
cat("   Seasonal variance:", round(air_var_seasonal, 3), "\n")
```

```
##    Seasonal variance: 1315.542
```

```
cat("   Residual variance:", round(air_var_residual, 3), "\n\n")
```

```
##    Residual variance: 374.056
```

```
cat("8. PERCENTAGE OF VARIANCE EXPLAINED:\n")
```

```
## 8. PERCENTAGE OF VARIANCE EXPLAINED:
```

```
cat("   Trend:", round((air_var_trend/air_var_total)*100, 1), "%\n")
```

```
##    Trend: 73.1 %
```

```
cat("   Seasonal:", round((air_var_seasonal/air_var_total)*100, 1), "%\n")
```

```
##    Seasonal: 9.1 %
```

```
cat("   Residual:", round((air_var_residual/air_var_total)*100, 1), "%\n\n")
```

```
##    Residual: 2.6 %
```

```
# Determine dominating component
cat("9. DOMINATING COMPONENT IDENTIFICATION:\n")
```

```
## 9. DOMINATING COMPONENT IDENTIFICATION:
```

```
if (air_var_trend > air_var_seasonal && air_var_trend > air_var_residual) {
  cat("   DOMINATING COMPONENT: TREND\n")
  cat("   Reason: Trend variance (", round((air_var_trend/air_var_total)*100, 1), "%) is highest\n", sep
} else if (air_var_seasonal > air_var_trend && air_var_seasonal > air_var_residual) {
  cat("   DOMINATING COMPONENT: SEASONAL\n")
} else {
  cat("   DOMINATING COMPONENT: RANDOM/RESIDUAL\n")
}
```

```
##    DOMINATING COMPONENT: TREND
##    Reason: Trend variance (73.1%) is highest
```

```
# ------------------------------------------------------------
# SUMMARY COMPARISON
# ------------------------------------------------------------
cat("\n\n", rep("=", 60), "\n", sep="")
```

```
##
##
## ============================================================
```

```r
cat("SUMMARY COMPARISON\n")
```

```
## SUMMARY COMPARISON
```

```r
cat(rep("=", 60), "\n\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

```r
cat("COMPARISON TABLE:\n")
```

```
## COMPARISON TABLE:
```

```r
cat(rep("-", 60), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("Dataset      | Total Var | Trend % | Seasonal % | Dominant Component\n")
```

```
## Dataset      | Total Var | Trend % | Seasonal % | Dominant Component
```

```r
cat(rep("-", 60), "\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("Nottem       | ", round(nottem_var_total, 1), " | ",
    round((nottem_var_trend/nottem_var_total)*100, 1), "% | ",
    round((nottem_var_seasonal/nottem_var_total)*100, 1), "% | SEASONAL\n", sep="")
```

```
## Nottem       | 73.5 | 1.1% | 93.4% | SEASONAL
```

```r
cat("AirPassengers| ", round(air_var_total, 1), " | ",
    round((air_var_trend/air_var_total)*100, 1), "% | ",
    round((air_var_seasonal/air_var_total)*100, 1), "% | TREND\n", sep="")
```

```
## AirPassengers| 14391.9 | 73.1% | 9.1% | TREND
```

```r
cat(rep("-", 60), "\n\n")
```

```
## - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

```r
cat("KEY OBSERVATIONS:\n")
```

```
## KEY OBSERVATIONS:
```

```r
cat("1. Nottem (Temperature Data):\n")
```

```
## 1. Nottem (Temperature Data):
```

```r
cat("   - Strong seasonal patterns (93.4% of variance)\n")
```

```
##    - Strong seasonal patterns (93.4% of variance)
```

```r
cat("   - Minimal trend (1.1% of variance)\n")
```

```
##    - Minimal trend (1.1% of variance)
```

```r
cat("   - Typical of climate/weather data with annual cycles\n\n")
```

```
##    - Typical of climate/weather data with annual cycles
```

```r
cat("2. AirPassengers (Economic Data):\n")
```

```
## 2. AirPassengers (Economic Data):
```

```r
cat("   - Strong upward trend (73.1% of variance)\n")
```

```
##    - Strong upward trend (73.1% of variance)
```

```r
cat("   - Moderate seasonal patterns (9.1% of variance)\n")
```

```
##    - Moderate seasonal patterns (9.1% of variance)
```

```r
cat("   - Reflects post-WWII growth in air travel industry\n\n")
```

```
##    - Reflects post-WWII growth in air travel industry
```

```r
cat("3. General Conclusion:\n")
```

```
## 3. General Conclusion:
```

```r
cat("   - Climate data → Seasonal dominance\n")
```

```
##    - Climate data → Seasonal dominance
```

```r
cat("   - Economic growth data → Trend dominance\n")
```

```
##    - Economic growth data → Trend dominance
```

```r
cat("   - Different domains show different dominant patterns\n")
```

```
##    - Different domains show different dominant patterns
```

```r
cat(rep("=", 60), "\n")
```

```
## = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =
```

Observations and Conclusions:

Key Findings:

1. **Nottem Dataset:**
   - **Dominant Component**: Seasonal (93.4% of variance)
   - **Interpretation**: Temperature data shows strong annual cycles with consistent seasonal patterns
   - **Trend**: Very weak (only 1.1%), indicating no significant long-term temperature change in this period
   - **Residual**: Small random variations (5.9%)
2. **AirPassengers Dataset:**
   - **Dominant Component**: Trend (73.1% of variance)
   - **Interpretation**: Strong upward trend showing rapid growth in air travel 1949-1960
   - **Seasonal**: Moderate seasonal patterns (9.1%) - higher travel in summer/ holidays
   - **Residual**: Very small random variations (2.6%)

Report Generated: 31 January 2026 **R Version:** R version 4.5.2 (2025-10-31 ucrt) **Packages Used:** ggplot2, forecast, ggfortify